

Deployment Documentation for ACB Backend Service

Requirements:

- **Operating System:** Ubuntu or similar linux/debian
- **Python:** 3.12+
- **PostgreSQL:** 17.9+
- **Node.js:** 24.x
- **PM2:** Process Manager

Installation and Setup

1. Update System Packages:

```
$ sudo apt update && sudo apt upgrade -y
```

2. Install Node.js and PM2:

```
$ curl -fsSL https://deb.nodesource.com/setup_24.x -o nodesource_setup.sh
$ sudo -E bash nodesource_setup.sh
$ sudo apt install -y nodejs
$ node -v
$ npm -v
$ sudo npm install pm2 -g
$ pm2 status
```

3. Install PostgreSQL & Create Database + Apps DB User:

```
$ sudo apt install -y postgresql-17 postgresql-contrib
$ sudo apt install -y postgresql-17-postgis-3
$ psql --version
$ sudo -u postgres psql

postgres=# CREATE USER acb_admin WITH PASSWORD 'your_secure_password';
postgres=# CREATE DATABASE acb_be OWNER acb_admin;
postgres=# GRANT ALL PRIVILEGES ON DATABASE acb_be TO acb_admin;
postgres=# \c acb_be;
acb_be=# CREATE EXTENSION postgis;
```

4. Configure PostgreSQL for Remote Access:

- Edit the PostgreSQL configuration file:

```
$ sudo nano /etc/postgresql/17/main/postgresql.conf
```

- Locate the 'listen_addresses' parameter and modify it as follows:

```
listen_addresses = '*'
```

- Edit the pg_hba.conf file to allow remote connections:

```
$ sudo nano /etc/postgresql/17/main/pg_hba.conf
```

- Add the following line at the end of the file:

```
host    acb_be    acb_admin    0.0.0.0/0    md5
```

- Restart PostgreSQL:

```
$ sudo systemctl restart postgresql
```

5. Clone Repository:

```
$ git clone https://github.com/WRI-Indonesia/acb-dashboard-be.git  
$ cd acb-dashboard-be
```

6. Configure Environment Variables:

```
$ nano .env  
  
DB_SQLALCHEMY_URI="postgresql+psycopg2://acb_admin:your_secure_password@localhost:5432/acb_be"  
NBSTOOLS_GWC_SERVICE_URL=https://nbstool.scenecoalition.org/layer/gwc/service/wmts  
NBSTOOLS_LEGEND_URL=https://nbstool.scenecoalition.org/layer/nbs/wms
```

7. Install Python PIP and Virtual Environment:

```
$ sudo apt install -y python3-pip python3-virtualenv libpq-dev
$ virtualenv venv
$ source venv/bin/activate
$ python -m pip install -r requirements.txt
```

8. Database Setup

```
$ flask db upgrade
```

9. Run Background Jobs Using PM2:

```
$ pm2 --name="acb-dashboard-be" start "python main.py"
$ pm2 status
$ pm2 log acb-dashboard-be
```

10. Setup PM2 Running When Startup

```
$ pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup
systemd -u someuser --hp /home/someuser
```

notes: need to copy the command from your terminal after "pm2 startup"

11. If Jobs not starting after reboot/startup

```
$ pm2 resurrect
```